

Implementing drought early warning systems in Africa:

policy lessons and future needs



Key policy Messages

- Current drought management practice in Africa focuses on mitigation and relief, and less on adaptation. A shift towards a risk based drought management approach offers opportunities to move away from expensive relief actions. This requires drought monitoring using suitable indicators, drought vulnerability assessment, seasonal meteorological and hydrological forecasts, a user friendly early warning, training and public awareness, and improved data sharing.
- Clear institutional responsibilities are prerequisite to effective drought forecasting and warning. Institutions involved in drought mitigation and adaptation should be connected to the formal frameworks concerning drought. A continuous and efficient collaboration of institutions and services involved is required and this should not only be limited to drought periods.
- Low level of technical and scientific personnel in most organisations issuing early warning products is a major constraint. Capacity is required at all levels (researchers, meteorologists, technology transfer, farmers, policy makers, communities, etc.) for effective interpretation and usage of forecasting and early warning products.
- Drought forecasts and warnings should be targeted to provide key information to users that they can take decisions on. Traditional knowledge can be integrated to supplement drought forecasts and early warning systems.
- Seasonal meteorological and hydrological forecasts have skill in predicting drought conditions in most regions across Africa; though this predictability varies across the continent, depending on the type of climate and scale.
- Very high resolution simulations of projected climate change show considerable variation across the continent. Subtropical and extra-tropical areas are expected to become dryer to much dryer. Eastern Africa and the Nile basin are expected to become wetter, while there is less confidence of the signal in Western Africa and the Sahel. In several areas an increase in heat stress is expected. The ability to forecast heat is, however, expected to remain high indicating drought forecasting and warning are effective in both current and future climates.

Introduction

Large parts of Africa are drought prone and are regularly faced with substantial economic losses and humanitarian suffering as a consequence of drought. The recent 2011 drought in the Horn of Africa is estimated to have caused up to a quarter of a million deaths, and left over 13 million people in the region of humanitarian aid. Over \$ 1.3 billion was spent on relief. The Final Declaration of the High Level Meeting on National Drought Policy (UN HMNDP) in Geneva (March 2013) encouraged all Governments around the world to develop and implement National Drought Management Policies. A key element in the recommendations was to promote greater collaboration to enhance the quality of preparedness for drought from local through to national and global levels. The recently completed DEWFORA research project, financed as a part of the EU 7w Framework Programme fosters such collaboration, thereby contributing to reducing vulnerability and strengthen preparedness to droughts across Africa.

Towards a protocol for developing early warning systems

An early warning system that supports the timely implementation of drought impact mitigation is one key element of an effective drought management strategy. Within DEWFORA a protocol for implementing early warning systems, based on a suite of integrated data, analyses and research on drought forecasting was developed. This protocol covers the whole chain from monitoring and vulnerability assessment, to forecasting, warning, and response and knowledge dissemination.

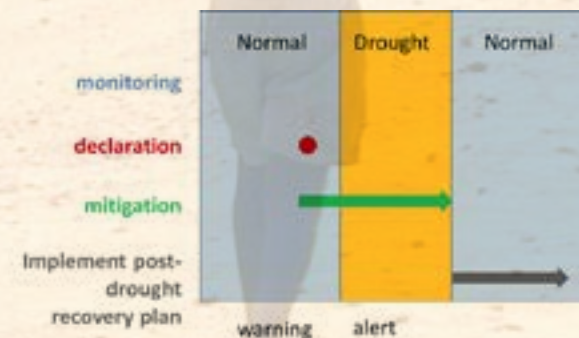
The protocol includes four steps to assess the scientific knowledge and the social capacity to use the knowledge in a specific country, basin or region. The protocol involves answering the following questions:

- What is the available science? Evaluating the monitoring system of impending drought. Evaluating the available drought forecasts. Definition of risk levels and analysing the signs of drought in an integrated vulnerability approach.
- What are the societal capacities? Evaluating the institutional framework that enables policy development.
- How can science be translated into policy? Linking science indicators into the actions/interventions that society needs to implement. Evaluation of policy implementation.
- How can society benefit from the forecast? Evaluating the provision of information to potentially affected groups.

Practical warning to avoid damage

Based on the assessment of drought vulnerability in a specific country or region we suggest thresholds for management actions to respond to drought forecasts and link predictive indicators to relevant potential mitigation strategies. The slow onset of droughts typically means that it is possible to determine vulnerability thresholds in advance that may trigger mitigation actions for different decision makers or stakeholders. The manager of a reservoir could change operational rules to ensure urban water supply, while a farmer could change the feeding plans for his or her livestock according to expected crop production.

DEWFORA gathered information on vulnerability so early warnings can be issued at sufficient lead time and drought mitigation planning can be implemented at an earlier stage. The idea is summarized in the figure below and entails to issue a drought warning before the effects of drought are manifested.



Case studies

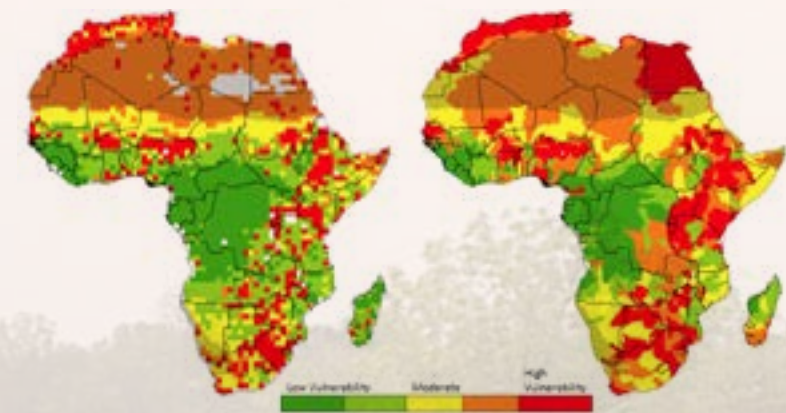
The DEWFORA project focused on four case studies, each with a specific focus area of the issues addressed in the project.

The case studies include (see Figure):

- the Limpopo basin
- the Oum er Rbia basin
- the Niger basin
- the Nile basin

Pan-African Drought Vulnerability

The DEWFORA team evaluated drought vulnerability of agricultural systems on a Pan-African scale (see figure below) in a gridded model (left map) and on a sub-basin scale (right map). The gridded model and the sub-basin approach show large similarities: high vulnerability the climatic dry north and south as well as in parts of the Rift valley and parts of West-Africa. In the desert regions (Sahara, Sahel, Namib Desert) precipitation is extremely low, and management of natural and social systems is not based on early warning systems but rather in permanent measures (dashed areas in Figures).



Improving mitigation and adaptation practices

Currently, the most common drought mitigation actions implemented in the basins include food aid (Limpopo, Nile), seed supply (Oum er Rbia, Nile) and harvesting of rainfall water (Limpopo, Nile). Other drought mitigation actions include the use of Indigenous Knowledge Systems (Limpopo, Oum er Rbia), education and public awareness campaigns (Nile, Limpopo). Adaptation actions in the basins include construction of water harvesting infrastructure (multi-purpose dams at basin scale in Niger basin), water harvesting and improved dry-land farming practices (Limpopo, Nile). Supplementary irrigation is key in the Oum er Rbia basin.

The need for strategies to manage and mitigate drought vulnerability is generally appreciated in all the basins. During the DEWFORA project, drought forecasting tools were developed and operationalized. For example in the Niger basin, weekly drought bulletins currently allow people in the area to make no-regret livelihood investments and changes in livelihood. However, many mitigation and adaptation actions are still hampered by several factors such as insufficient technical and scientific personnel and inadequate financial resources for adequate gauging stations, data collection, infrastructure and technology for advanced models and forecasting systems, and drought vulnerability analyses. Further, different institutions could better coordinate their activities so that measures become more effective.

Overall, the results from the DEWFORA project highlight that whilst early warning, mitigation and adaptation in the African region has been tackled in several ways, for it to be effective, there is need to develop human resources involved in designing early warning systems and also to incorporate local knowledge. A discussion platform needs to be created to ensure effective communication between all the concerned stakeholders.



Location of case study areas in Africa

What is the science available?

What are the societal capacities?

How can science be translated into policy?

How can society benefit from the forecast?

The DEWFORA project

The main aim of the DEWFORA project is to develop a framework to reduce vulnerability and strengthen preparedness to droughts in Africa by advancing drought forecasting, early warning and mitigation practices. The framework addresses monitoring, predicting, timely warning and response to droughts at the seasonal time scale, applicable within the institutional context of African countries. The project contributes through improved methods for identification of vulnerable regions taking into account the increased hazard due to climate change, and feasible adaptation measures.

Starting/Ending date of project:

1th of January 2010 - 31th December 2013

Type of R&D:

- Collaborative project
- Small or medium scale focused research project

Programme:

7th Framework Programme

Theme 6: Environment (including climate change)



Selected related activities:

- EU Framework Projects:
AFROMAISON, HEALTHY FUTURES, CLUVA, CLIMAFRICA, AfriCAN CLIMATE, AFRICA-GHG, CLARA, GLOWASIS, CIRCE.
- Drought initiatives:
CIS-SPI, Drought R&SPI, GDIS, GEO community & GEOSS, NeWater project.
- Websites on drought in Africa:
<http://www.unicefusa.org/work/emergencies/horn-of-africa/>
http://en.wikipedia.org/wiki/2011_East_Africa_drought

Participating countries/institutes

	Deltares (coordinator) <i>Netherlands</i>
	Council for Scientific and Industrial Research <i>South Africa</i>
	Dinder Center for Environmental Research <i>Sudan</i>
	European Centre for Medium-range Weather Forecasts <i>Europe</i>
	German Research Centre for Geosciences <i>Germany</i>
	Hydraulic Research Institute - Nile Basin Capacity Building Network for River Engineering <i>Egypt</i>
	Joint Research Centre <i>Europe</i>
	Mediterranean Agronomic Institute of Zaragoza <i>Spain</i>
	IGAD Climate Prediction and Applications Centre <i>Kenya</i>
	Nile Forecast Center <i>Egypt</i>
	Potsdam Institute for Climate Impact Research <i>Germany</i>
	Institut Agronomique et Vétérinaire Hassan II <i>Morocco</i>
	UNESCO-IHE Institute for Water Education <i>Netherlands</i>
	Universidad Politécnica de Madrid <i>Spain</i>
	University Eduardo Mondlane, Faculty of Engineering <i>Mozambique</i>
	University of Porto, Faculty of Engineering <i>Portugal</i>
	WR Nyabeze & Associates <i>South Africa</i>
	WaterNet Trust <i>Botswana</i>
	Wetlands International – Sahelian Sub Regional Office <i>Mali</i>

Web-Links DEWFORA:

www.dewfora.net

<http://edo.jrc.ec.europa.eu/dewfora/php/index.php?id=4119>

www.euronews.com/2013/06/17/africa-is-always-at-risk-of-drought